

JetBrains Lab – Incident Response Report

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Platform: CyberDefenders

Category: Network Forensics

Executive Summary

This investigation analyzed network traffic captured from a compromised TeamCity server. The attacker (23.158.56.196) exploited the application, enumerated REST API endpoints, uploaded a malicious ZIP plugin (NSt8bHTg.zip), and deployed a JSP web shell capable of executing operating system commands through Java ProcessBuilder.

Objectives

- Identify the attacker.
- Analyze HTTP traffic.
- Reconstruct the attack timeline.
- Extract Indicators of Compromise (IOCs).
- Map activity to MITRE ATT&CK.

Tools Used

- Wireshark

Investigation Methodology

1. Open the PCAP in Wireshark.
2. Filter HTTP traffic.
3. Review authentication requests.
4. Inspect TeamCity REST API requests.
5. Follow HTTP streams.
6. Identify uploaded files.
7. Analyze the JSP web shell.
8. Document IOCs.

Evidence & Screenshot Placement

Initial HTTP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
23913	755.103371	23.158.56.196	172.17.0.2	HTTP	608	GET /login.html HTTP/1.1
23983	764.292866	23.158.56.196	172.31.25.119	HTTP	1388	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
23984	764.292896	23.158.56.196	172.17.0.2	HTTP	1388	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24034	784.825107	23.158.56.196	172.31.25.119	HTTP	1388	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24034	784.825139	23.158.56.196	172.17.0.2	HTTP	1388	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24169	859.451126	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24170	859.451132	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24200	866.987237	23.158.56.196	172.31.25.119	HTTP	1396	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24201	866.987286	23.158.56.196	172.17.0.2	HTTP	1396	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24230	873.215463	23.158.56.196	172.31.25.119	HTTP	1132	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24237	873.215494	23.158.56.196	172.17.0.2	HTTP	1132	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24412	958.003913	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24413	958.003920	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24472	975.811843	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24473	975.811876	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24491	979.038105	23.158.56.196	172.31.25.119	HTTP	1132	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24492	979.038148	23.158.56.196	172.17.0.2	HTTP	1132	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24506	981.666080	23.158.56.196	172.31.25.119	HTTP	1127	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24507	981.666113	23.158.56.196	172.17.0.2	HTTP	1127	POST /loginSubmit.html HTTP/1.1 (application/x-www-form-urlencoded)
24666	1046.946329	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24667	1046.946336	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
24701	1054.157125	23.158.56.196	172.31.25.119	HTTP	345	GET /hax?jsp=app/rest/server;.jsp HTTP/1.1
24702	1054.157131	23.158.56.196	172.17.0.2	HTTP	345	GET /hax?jsp=app/rest/server;.jsp HTTP/1.1
24721	1054.444574	23.158.56.196	172.31.25.119	HTTP/1.1	218	POST /hax?jsp=app/rest/users;.jsp HTTP/1.1, JSON (application/json)
24722	1054.444599	23.158.56.196	172.17.0.2	HTTP/1.1	218	POST /hax?jsp=app/rest/users;.jsp HTTP/1.1, JSON (application/json)
24733	1054.916606	23.158.56.196	172.31.25.119	HTTP	387	POST /hax?jsp=app/rest/users/id:2/tokens/md5r0yemB0;.jsp HTTP/1.1
24734	1054.916638	23.158.56.196	172.17.0.2	HTTP	387	POST /hax?jsp=app/rest/users/id:2/tokens/md5r0yemB0;.jsp HTTP/1.1
24748	1054.211169	23.158.56.196	172.31.25.119	HTTP	494	POST /authenticationTest.html?csrf HTTP/1.1

Frame 23912: 608 bytes on wire (4864 bits), 608 bytes captured (4864 bits)
 Linux cooked capture v2
 Internet Protocol Version 4, Src: 23.158.56.196, Dst: 172.17.0.2
 Transmission Control Protocol, Src Port: 57442, Dst Port: 8111, Seq: 1, Ack: 1, Len: 536
 Hypertext Transfer Protocol: Protocol

Packets: 33279 · Displayed: 400 (1.2%) Profile: Default

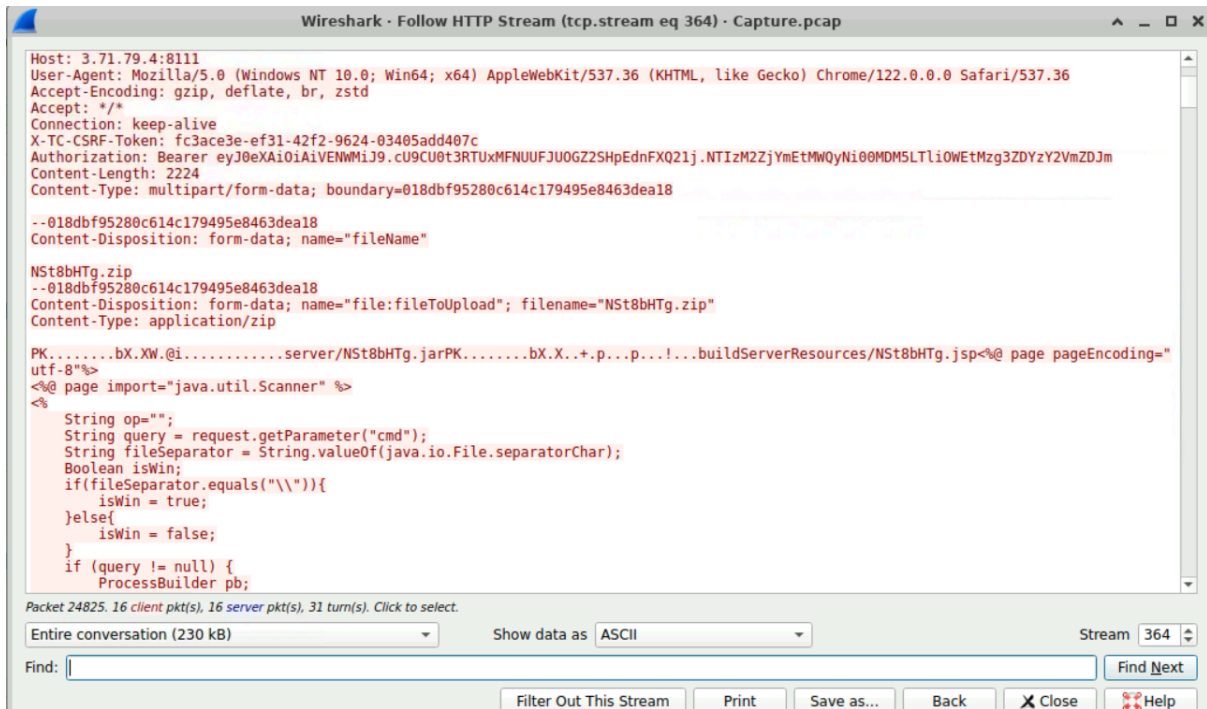
TeamCity Enumeration

No.	Time	Source	Destination	Protocol	Length	Info
24702	1054.157131	23.158.56.196	172.17.0.2	HTTP	345	GET /hax?jsp=app/rest/server;.jsp HTTP/1.1
24721	1054.444574	23.158.56.196	172.31.25.119	HTTP/1.1	218	POST /hax?jsp=app/rest/users;.jsp HTTP/1.1, JSON (application/json)
24722	1054.444599	23.158.56.196	172.17.0.2	HTTP/1.1	218	POST /hax?jsp=app/rest/users;.jsp HTTP/1.1, JSON (application/json)
24733	1054.916606	23.158.56.196	172.31.25.119	HTTP	387	POST /hax?jsp=app/rest/users/id:2/tokens/md5r0yemB0;.jsp HTTP/1.1
24734	1054.916638	23.158.56.196	172.17.0.2	HTTP	387	POST /hax?jsp=app/rest/users/id:2/tokens/md5r0yemB0;.jsp HTTP/1.1
24748	1056.211169	23.158.56.196	172.31.25.119	HTTP	494	POST /authenticationTest.html?csrf HTTP/1.1
24749	1056.211201	23.158.56.196	172.17.0.2	HTTP	494	POST /authenticationTest.html?csrf HTTP/1.1
24760	1056.422781	23.158.56.196	172.31.25.119	HTTP	420	GET /hax?jsp=app/rest/debug/jvm/systemProperties;.jsp HTTP/1.1
24761	1056.422814	23.158.56.196	172.17.0.2	HTTP	420	GET /hax?jsp=app/rest/debug/jvm/systemProperties;.jsp HTTP/1.1
24825	1071.315015	23.158.56.196	172.31.25.119	HTTP	848	POST /admin/pluginUpload.html HTTP/1.1 (application/zip)
24826	1071.315047	23.158.56.196	172.17.0.2	HTTP	848	POST /admin/pluginUpload.html HTTP/1.1 (application/zip)
24840	1071.578220	23.158.56.196	172.31.25.119	HTTP	596	GET /admin/admin.html?item=plugins HTTP/1.1
24841	1071.578252	23.158.56.196	172.17.0.2	HTTP	596	GET /admin/admin.html?item=plugins HTTP/1.1
25370	1073.345245	23.158.56.196	172.31.25.119	HTTP	144	POST /admin/plugins.html HTTP/1.1 (application/x-www-form-urlencoded)
25371	1073.345289	23.158.56.196	172.17.0.2	HTTP	144	POST /admin/plugins.html HTTP/1.1 (application/x-www-form-urlencoded)
25510	1117.226064	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
25511	1117.226071	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
25536	1128.649175	23.158.56.196	172.31.25.119	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1
25537	1128.649206	23.158.56.196	172.17.0.2	HTTP	681	GET /authenticationTest.html?csrf HTTP/1.1

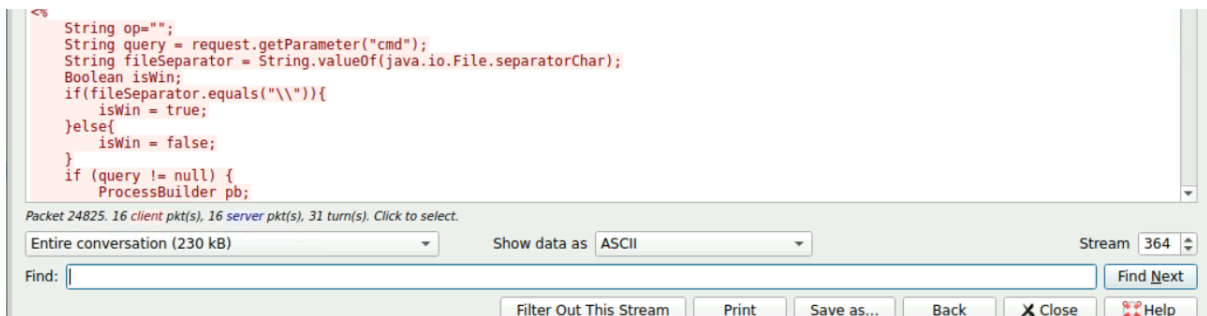
Frame 24702: 345 bytes on wire (2760 bits), 345 bytes captured (2760 bits)
 Linux cooked capture v2
 Internet Protocol Version 4, Src: 23.158.56.196, Dst: 172.17.0.2
 Transmission Control Protocol, Src Port: 54144, Dst Port: 8111, Seq: 1, Ack: 1, Len: 273
 Hypertext Transfer Protocol: Protocol

Packets: 33279 · Displayed: 400 (1.2%) Profile: Default

Malicious Plugin Upload



Web Shell Analysis



Indicators of Compromise (IOCs)

IOC	Value
Attacker IP	23.158.56.196
Victim IP	172.31.25.119
Port	8111
Uploaded ZIP	NSt8bHTg.zip
Web Shell	NSt8bHTg.jsp
Upload Endpoint	/admin/pluginUpload.html

MITRE ATT&CK

- T1190 - Exploit Public-Facing Application
- T1059 - Command and Scripting Interpreter
- T1505.003 - Web Shell
- T1071 - Application Layer Protocol
- T1083 - File and Directory Discovery

Conclusion

The investigation confirmed successful exploitation of the TeamCity server, upload of a malicious plugin, deployment of a JSP web shell, and remote command execution. The collected IOCs can be used to improve detection and future incident response.